

ARASH VASHAGH

✉ Email

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🌐 LinkedIn

🎓 Google Scholar

🌐 Portfolio

Summary

Machine Learning Developer completing an MSc in computer science with research focus on adversarial machine learning. Published researcher with several certifications and hands-on projects across AI, cybersecurity, and cloud computing. Available for full-time work and open to relocation across Canada for on-site work.

Skills

Programming: Python, Java, Kotlin, JavaScript, SQL, C++

AI/ML: PyTorch, TensorFlow, Agentic Design, RAG/LLMOps, Scikit-learn, Reinforcement Learning, Stable-Baselines3, Gymnasium, NLP

Web, Backend & APIs: React, HTML, CSS, Streamlit, FastAPI, Flask, REST APIs, OpenAI API, JSON Processing

Cloud & Deployment: AWS, Microsoft Azure, Google Cloud, Streamlit Community Cloud, Docker

Tools & Libraries: NumPy, Pandas, Matplotlib, PIL, Jupyter, VS Code, IntelliJ IDEA, Postman, PgAdmin 4, GitHub, MLflow

Languages: Persian (Native), English (Professional)

Certifications

- **AI and Machine Learning:** AWS Certified AI Practitioner; Microsoft AI-900; DeepLearning.AI Agentic AI
- **Cybersecurity:** CompTIA Security+; Microsoft SC-900
- **Cloud Computing:** AWS Certified Cloud Practitioner; Microsoft AZ-900; Google Cloud Digital Leader (results pending)
- **Data and Project Management:** CompTIA Data+; Microsoft DP-900; CompTIA Project Management Essentials; CompTIA DataSys+ (results pending)

Experience

Graduate Research Assistant / Team Lead — UNB — Fredericton, Canada

Sep 2024 – Present

- Conducting MSc thesis research in Adversarial Machine Learning, focusing on attacks, detection, and trustworthy AI.
- Co-authored a [survey on adversarial attacks](#) and another [survey on adversarial defense mechanisms](#).
- Proposed and implemented two adversarial detection mechanisms, currently under internal faculty review.
- Developing an adversarial attack on fusion models, with manuscript in progress.
- Coordinating graduate student work and managing the [Trustworthy and Secure AI \(TSAI\) LinkedIn page](#) for lab updates and educational content.

Graduate Teaching Assistant — UNB — Fredericton, Canada

Jan 2025 – Apr 2026

- *Software Engineering* (Winter 2026) – Java
- *Data Communication and Network Modeling* (Fall 2025) – Python
- *Data Science for Big Data Analytics* (Winter 2025) – Python

Conference Reviewer — Remote

Jul 2024 – Oct 2024

- Reviewed manuscripts and provided structured feedback at international conferences:
 - *ArtInHCI 2024* – Artificial Intelligence and Human-Computer Interaction (Kunming, China) [\[Certificate\]](#)
 - *BIBE 2024* – Biological Information and Biomedical Engineering (Hohhot, China) [\[Certificate\]](#)

Research Assistant — IUT — Isfahan, Iran

Feb 2021 – Feb 2024

- Conducted research on machine learning and healthcare, resulting in two IEEE publications [\[1\]](#) and [\[2\]](#).
- Collaborated with faculty on implementation, paper writing, and presenting at conferences.

Teaching Assistant — IUT — Isfahan, Iran

Feb 2021 – Feb 2024

- Courses: Artificial Intelligence, Embedded Systems, Data Structures, and Programming Labs in C++, Java, and Python.
- Gave tutorials and graded assignments.

Education

Master's Candidate — University of New Brunswick (UNB), Fredericton, Canada

Sep 2024 – Aug 2026

- Awarded Graduate Scholarship and Research Grant.
- Major: Computer Science
- Research Focus: Adversarial Machine Learning

Bachelor of Science — Isfahan University of Technology (IUT), Isfahan, Iran

Sep 2019 – Feb 2024

- Major: Computer Engineering
- Minor: Intelligent Systems

Selected Projects

PaperWise AI: Deployed Agentic Research Paper RAG / LLMOps App

Jun 2026

- Built and deployed a public Streamlit app for research paper question answering.
- Implemented PDF ingestion, text extraction, chunking, TF-IDF retrieval, OpenAI API integration, and offline fallback.
- Applied agentic design through suggested questions, reviewer critique mode, structured paper summaries, multi-paper comparison, and PDF answer export.
- Displayed RAG pipeline metrics including retrieval scores, response latency, retrieved chunks, and citation usage.
- Technologies: Python, Streamlit, Scikit-learn, OpenAI API, pypdf, ReportLab, Streamlit Community Cloud.
- Live app: [Streamlit](#).
- Source code: [GitHub](#).

Atrial Fibrillation Detection with Diffusion-Based Data Augmentation

2023

- Developed a deep learning pipeline for atrial fibrillation (AF) detection using cardiac signal preprocessing.
- Extracted R-R interval features and transformed 1-D ECG-derived data into 2-D Poincaré images for classification.
- Applied data augmentation using a diffusion model to improve the balance of dataset.
- Trained and evaluated CNN-based classifiers for AF detection using generated and preprocessed medical data.
- Published as an [IEEE conference paper](#)
- Technologies: Python, PyTorch, CNNs, DDPM, NumPy, MATLAB, biomedical signal processing, medical image-style preprocessing.
- Source code: [GitHub](#).

PromptShield: Adversarial Detection for Vision-Language Models

Jan 2026 – May 2026

- Developed a machine learning framework to detect adversarial images in vision-language models.
- Evaluated detection performance across multiple datasets, model backbones, and adversarial attack settings.
- Designed experiments for robustness evaluation, attack transferability analysis, and detector performance comparison.
- Built reproducible ML experiments for model training, evaluation, metric tracking, and result analysis.
- Technologies: Python, PyTorch, OpenCLIP, NumPy, Scikit-learn, adversarial machine learning.
- Manuscript and code under review at [NeurIPS 2026](#).
- Source code: [GitHub](#).

Voluntary Educator (AI Security)

AI Security and Adversarial Machine Learning (AML) Course — Online

Jun 2026 – Present

- Creating a Persian technical course on AI Security and AML to support learners facing language barriers.
- Developed slides and scripts on machine learning foundations for AML.
- Published course videos through a structured [YouTube playlist](#) and shared course materials on [GitHub](#).